# A Barvinok–Pataki Theorem for Archimedean Order Unit Spaces

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**Abstract**

We prove a generalisation of the Barvinok–Pataki rank bound to Archimedean order unit spaces equipped with a pointed cone. The classical result bounds the rank of extreme points of spectrahedra (feasible sets of semidefinite programs); our generalisation bounds the face dimension of feasible points under affine constraints in an arbitrary pointed cone, including the infinite-dimensional case. The face-dimension bound $d_F \leq m$ uses only pointedness of the cone and rank–nullity; finite-dimensionality of $d_F$ in infinite dimensions follows from a result of Weis–Shirokov that requires only a real vector space and a convex set. When specialised to C*-algebras via the Alfsen–Shultz face–commutant correspondence, this recovers the operator-algebraic Barvinok–Pataki bound $\sum n_j^2 \leq m + 1$.

## 1 Preliminaries

**Definition 1** (Ordered vector space with cone)**.** Let $V$ be a real vector space. A *cone* is a subset $V^+ \subseteq V$ that is closed under addition and non-negative scalar multiplication. The pair $(V, V^+)$ is an *ordered vector space* with partial order $x \leq y$ iff $y - x \in V^+$.

**Definition 2** (Pointed cone)**.** A cone $V^+$ is *pointed* (or *proper*) if $V^+ \cap (-V^+) = \{0\}$, i.e. the only element $v$ with both $v \geq 0$ and $-v \geq 0$ is $v = 0$.

**Definition 3** (Order unit and Archimedean property)**.** An element $e \in V^+$ is an *order unit* if for every $v \in V$ there exists $n \in \mathbb{N}$ such that $ne - v \in V^+$ (i.e. $v \leq ne$). The cone $V^+$ is *Archimedean* if, whenever $ne + v \in V^+$ for all $n \in \mathbb{N}$, it follows that $v \in V^+$. The triple $(V, V^+, e)$ is called an *Archimedean order unit space* (AOU space) if $V^+$ is a pointed Archimedean cone and $e$ is an order unit.

**Definition 4** (Face of a cone)**.** Let $V^+$ be a cone in $V$ and $x \in V^+$. The *face of* $V^+$ *generated by* $x$ is

$$\mathrm{face}_{V^+}(x) \;=\; \left\{\, w \in V^+ \;\middle|\; \exists\, \lambda > 0 \colon x - \lambda w \in V^+ \,\right\}.$$

Equivalently, $\mathrm{face}_{V^+}(x)$ consists of all elements of $V^+$ that can be "majorised" by a positive multiple of $x$. This is the smallest face of $V^+$ containing $x$ in its relative interior.

**Remark 5.** In a pointed cone, $\mathrm{face}_{V^+}(x) = \{0\}$ if and only if $x = 0$. For $x \neq 0$, the face is at least one-dimensional (it contains the ray $\{\lambda x : \lambda \geq 0\}$). For the PSD cone $\mathrm{PSD}_n$, the face generated by a rank-$r$ matrix $X$ is isomorphic to $\mathrm{PSD}_r$, with $\dim(\mathrm{face}(X)) = \frac{r(r+1)}{2}$ (real symmetric case) or $r^2$ (complex Hermitian case).

## 2 The Classical Barvinok–Pataki Bound

We first recall the classical proof for PSD matrices, as it motivates each step of the generalisation.

**Theorem 6** (Barvinok–Pataki, PSD case)**.** *Let $A_1, \dots, A_m \in \mathrm{Sym}_n(\mathbb{R})$ and $\beta_1, \dots, \beta_m \in \mathbb{R}$. Consider the spectrahedron*

$$C = \{\, X \in \mathrm{PSD}_n \mid \langle A_j, X\rangle = \beta_j,\ j = 1, \dots, m \,\}.$$

*If $X \in C$ is an extreme point of $C$, then $\frac{r(r+1)}{2} \le m$ where $r = \mathrm{rank}(X)$.*

*Proof.* Write $X = VBV^\top$ with $V \in \mathbb{R}^{n\times r}$, $B \in \mathrm{PSD}_r$ strictly positive definite. Define the compressed constraints $\hat{A}_j = V^\top A_j V \in \mathrm{Sym}_r$ and the linear map

$$\varphi\colon \mathrm{Sym}_r \to \mathbb{R}^m, \qquad [\varphi(D)]_j = \langle \hat{A}_j, D\rangle.$$

By rank–nullity, $\dim(\ker\varphi) \ge \frac{r(r+1)}{2} - m$. If $\ker\varphi \ne \{0\}$, choose $D \ne 0$ in $\ker\varphi$. Then $B(t) = B + tD$ satisfies all constraints for every $t$.

Since $B \succ 0$ and $D$ is a non-zero symmetric matrix with at least one negative eigenvalue (replacing $D$ by $-D$ if necessary), the map $t \mapsto \lambda_{\min}(B + tD)$ is continuous, positive at $t = 0$, and decreasing for $t > 0$. Let $t_0 > 0$ be the first value where $\lambda_{\min}(B + t_0 D) = 0$. Then $B(t_0) \succeq 0$ with $\mathrm{rank}(B(t_0)) < r$, and $VB(t_0)V^\top \in C$.

This contradicts $X$ being an extreme point of $C$ (since $X$ is a non-trivial convex combination of $VB(t_0)V^\top$ and $VB(-t_0')V^\top$ for suitably chosen $t_0'$), unless $\ker\varphi = \{0\}$, which forces $\frac{r(r+1)}{2} \le m$. □

## 3 Generalisation to Pointed Cones

The key observation is that the proof above uses only three ingredients:

1. **Rank–nullity** for the constraint map restricted to the face;
2. **Relative interior perturbation**: any kernel direction $D \in \mathrm{span}(F)$ can perturb $x$ in both directions while remaining in the face (since $x$ lies in the relative interior of $F$);
3. **Finite-dimensionality** of the face, to apply rank–nullity with finite dimensions.

Pointedness of the cone is used implicitly: it ensures that the face structure of $V^+$ is non-trivial (every face is a proper sub-cone).

**Theorem 7** (Barvinok–Pataki for pointed cones)**.** *Let $V$ be a real vector space, $V^+ \subseteq V$ a pointed cone, and $\alpha_1, \dots, \alpha_m \in V^*$ linear functionals. Define*

$$C = \{\, x \in V^+ \mid \alpha_j(x) = b_j,\ j = 1, \dots, m \,\}$$

*for given $b_1, \dots, b_m \in \mathbb{R}$. Let $x \in C$ and let $F = \mathrm{face}_{V^+}(x)$ be the face of $V^+$ generated by $x$. Assume that $d_F := \dim(\mathrm{span}(F))$ is finite. Define the restricted constraint map*

$$\varphi_F\colon \mathrm{span}(F) \to \mathbb{R}^m, \qquad [\varphi_F(v)]_j = \alpha_j(v).$$

*If $x$ is an extreme point of $C$, then*

$$\boxed{d_F \le m.}$$

*Equivalently, $\dim(F) \le m$ (where $\dim(F) = \dim(\mathrm{aff}(F)) = d_F$, since $0 \in F$ for every face of a cone).*

*Proof.* By rank–nullity applied to $\varphi_F$,

$$d_F \;=\; \dim(\ker\varphi_F) + \dim(\operatorname{im}\varphi_F) \;\le\; \dim(\ker\varphi_F) + m.$$

It suffices to show that $\ker\varphi_F = \{0\}$ when $x$ is extreme in $C$. Suppose for contradiction that there exists $D \in \ker\varphi_F$, $D \neq 0$.

**Step 1: The perturbation preserves feasibility.** Since $D \in \operatorname{span}(F) \cap \ker\varphi_F$, we have $\alpha_j(D) = 0$ for all $j$. Therefore, for any $t \in \mathbb{R}$:

$$\alpha_j(x + tD) = \alpha_j(x) + t\,\alpha_j(D) = b_j.$$

So $x + tD$ satisfies all affine constraints whenever it lies in $V^+$.

**Step 2: The point $x$ lies in the algebraic relative interior of $F$.** The face $F = \operatorname{face}_{V^+}(x)$ is by definition the smallest face of $V^+$ containing $x$ in its relative interior. In particular, for every $D \in \operatorname{span}(F)$ there exists $\varepsilon > 0$ such that $x + tD \in F \subseteq V^+$ for all $|t| < \varepsilon$.

To see this explicitly: write $D = w_1 - w_2$ with $w_1, w_2 \in F$. By definition of $F$, there exist $\lambda_1, \lambda_2 > 0$ with $x - \lambda_i w_i \in V^+$. For $0 < t < \min(\lambda_1, \lambda_2)$,

$$x + tD = \underbrace{(x - tw_2)}_{\in V^+} + \underbrace{tw_1}_{\in V^+} \;\in\; V^+,$$

where $x - tw_2 = (1 - t/\lambda_2)\,x + (t/\lambda_2)(x - \lambda_2 w_2) \in V^+$ by convexity. The same argument applies to $x - tD$.

**Step 3: Contradiction with extremality.** Combining Steps 1 and 2: for sufficiently small $\varepsilon > 0$, both $x^+ := x + \varepsilon D$ and $x^- := x - \varepsilon D$ lie in $F \subseteq V^+$ and satisfy all constraints, hence $x^+, x^- \in C$. Since $D \neq 0$, they are distinct ($x^+ - x^- = 2\varepsilon D \neq 0$). Then

$$x \;=\; \tfrac{1}{2}\,x^+ \;+\; \tfrac{1}{2}\,x^-$$

is a non-trivial convex combination of two distinct elements of $C$, contradicting the extremality of $x$.

Therefore $\ker\varphi_F = \{0\}$, and we conclude $d_F \le m$. □

# 4 Application to C*-Algebras and AOU Spaces

**Corollary 8** (Barvinok–Pataki for AOU spaces)**.** *Let $(V, V^+, e)$ be an Archimedean order unit space with $\dim(V) < \infty$. Let $\alpha_1, \ldots, \alpha_m \in V^*$ and consider the state-like set*

$$C = \big\{\, x \in V^+ \;\big|\; \alpha_j(x) = b_j,\ j = 1, \ldots, m \,\big\}.$$

*For any extreme point $x$ of $C$, the face of $V^+$ generated by $x$ satisfies $d_F = \dim(\operatorname{span}(\operatorname{face}_{V^+}(x))) \le m$.*

*Proof.* By definition, an AOU space has a pointed cone. The Archimedean property implies $V^+$ is closed in the order-unit norm. In finite dimensions, every face is finite-dimensional. Apply Theorem 7. □

**Corollary 9** (Barvinok–Pataki for C*-algebras)**.** *Let $A$ be a unital C*-algebra and $a_1, \ldots, a_m \in A_{\mathrm{sa}}$. Consider the affinely constrained state space*

$$C = \big\{\, \varphi \in S(A) \;\big|\; \varphi(a_j) = \beta_j,\ j = 1, \ldots, m \,\big\},$$

*where $S(A) = \{\varphi \in A^* : \varphi \ge 0,\ \varphi(\mathbf{1}) = 1\}$. If $\varphi$ is an extreme point of $C$, then*

$$\dim_{\mathbb{R}}\big(\pi_\varphi(A)'\big)_{\mathrm{sa}} \;\le\; m + 1,$$

*where $\pi_\varphi$ is the GNS representation of $\varphi$. Equivalently, if $\pi_\varphi(A)' \cong \bigoplus_j M_{n_j}(\mathbb{C})$ (Artin–Wedderburn), then $\sum_j n_j^2 \le m + 1$.*

*Proof.* The state space $S(A)$ is a base for the cone $A_+^*$ (positive functionals), cut out by the normalisation constraint $\varphi(\mathbf{1}) = 1$. The total number of affine constraints on $A_+^*$ is $m+1$ (the $m$ given constraints plus normalisation).

By the Alfsen–Shultz face–commutant correspondence, the face of $S(A)$ generated by $\varphi$ is affinely isomorphic to the state space $S(\pi_\varphi(A)')$ of the commutant. In particular,

$$\dim\bigl(\mathrm{face}_{S(A)}(\varphi)\bigr) = \dim_{\mathbb{R}}\bigl(\pi_\varphi(A)'\bigr)_{\mathrm{sa}} - 1.$$

The cone $A_+^*$ is pointed (if $\varphi$ and $-\varphi$ are both positive, then $\varphi(a^*a) = 0$ for all $a$, so $\varphi = 0$) and weak-$*$ closed. When $A$ is infinite-dimensional, the finiteness of $d_F$ follows from Corollary 11 below (applied with $\ell = m+1$). Theorem 7 then gives $\dim(\mathrm{span}(\mathrm{face}(\varphi))) \leq m+1$, hence $\dim_{\mathbb{R}}(\pi_\varphi(A)')_{\mathrm{sa}} \leq m+1$. □

## 5 The Infinite-Dimensional Case and the Role of $d_F$

Theorem 7 assumes that $d_F = \dim(\mathrm{span}(F))$ is finite. In infinite dimensions this is a non-trivial hypothesis: the face of a cone generated by a point can be infinite-dimensional. We now show that this hypothesis is *automatically satisfied* whenever the number of constraints is finite, using a result of Weis–Shirokov that is purely convex-geometric.

**Theorem 10** (Weis–Shirokov [4, Theorem 3.3])**.** *Let $V$ be a real vector space, $K \subseteq V$ a convex set, and $f\colon K \to \mathbb{R} \cup \{+\infty\}$ a generalised affine map. Let $C$ denote a sublevel set $\{x \in K : f(x) \leq \alpha\}$ or a level set $\{x \in K : f(x) = \alpha\}$, and let $x \in C$. If the face $F_C(x)$ of $C$ generated by $x$ has dimension $m \in \mathbb{N}_0$, then the face $F_K(x)$ of $K$ generated by $x$ has dimension $m$ or $m+1$.*

The multi-constraint version follows by induction (Corollary 5 of [4]): if $C$ is the intersection of $\ell$ sublevel and/or level sets of generalised affine maps, and $F_C(x)$ has dimension $m$, then $\dim(F_K(x)) \in \{m, m+1, \ldots, m+\ell\}$.

**Corollary 11** (Finiteness of $d_F$)**.** *Let $K$ be a convex set in a real vector space $V$, and let $C$ be defined by $\ell$ generalised affine constraints on $K$. If $x$ is an extreme point of $C$, then*

$$\dim\bigl(F_K(x)\bigr) \;\leq\; \ell.$$

*In particular, the face of $K$ generated by any extreme point of $C$ is finite-dimensional, regardless of* $\dim(V)$.

*Proof.* Since $x$ is extreme in $C$, the face $F_C(x) = \{x\}$ has dimension 0. By the multi-constraint version of Theorem 10, $\dim(F_K(x)) \in \{0, 1, \ldots, \ell\}$. □

**Remark 12** (No operator-algebraic assumptions needed)**.** Theorem 10 and Corollary 11 require only a real vector space $V$ and a convex subset $K \subseteq V$. In particular:

- No cone structure, pointedness, or Archimedean property is needed to establish that $d_F$ is finite.
- No C*-algebraic or operator-algebraic notions are needed.
- The operator-algebraic machinery (GNS representation, face–commutant correspondence, Artin–Wedderburn decomposition) is used only in a second step: to *interpret* the finite-dimensional face algebraically as a commutant and extract the multiplicity bound $\sum n_j^2 \leq m+1$.

This clarifies the logical structure of the infinite-dimensional case: Weis–Shirokov provides the purely geometric bound on $d_F$, and the Alfsen–Shultz correspondence translates it into algebra.

Combining Corollary 11 with Theorem 7, we obtain the following infinite-dimensional version of the main theorem that does not require an a priori finiteness assumption on $d_F$.

**Corollary 13** (Barvinok–Pataki for infinite-dimensional pointed cones)**.** *Let $V$ be a (possibly infinite-dimensional) real vector space, $V^+ \subseteq V$ a pointed closed cone, and $\alpha_1, \ldots, \alpha_m \in V^*$ linear functionals. Define $C = \{x \in V^+ : \alpha_j(x) = b_j\}$. If $x$ is an extreme point of $C$, then*

$$d_F = \dim\big(\mathrm{span}(\mathrm{face}_{V^+}(x))\big) \leq m.$$

*Proof.* By Corollary 11 (with $K = V^+$ viewed as a convex set and $\ell = m$), $d_F$ is finite. Theorem 7 then gives $d_F \leq m$. □

# 6 Special Cases Recovered

1. **Classical Barvinok–Pataki** ($A = M_n(\mathbb{C})$). The unique irreducible representation is the identity, so $\pi_\varphi(A)' \cong M_r(\mathbb{C})$ where $r = \mathrm{rank}(\rho_\varphi)$. Hence $r^2 \leq m + 1$, or $r^2 \leq m$ when one of the constraints is the trace normalisation already built into the spectrahedron.

2. **Carathéodory's theorem** ($A = C(X)$, abelian). All irreducible representations are one-dimensional, so the commutant decomposes as $\bigoplus_{j=1}^k \mathbb{C}$. The bound gives $k \leq m + 1$: extreme measures under $m$ moment constraints are supported on at most $m + 1$ points.

3. **Block-diagonal algebras** ($A = \bigoplus_i M_{d_i}(\mathbb{C})$). The bound $\sum_i r_i^2 \leq m + 1$ simultaneously constrains ranks within blocks *and* the number of active blocks.

4. **Infinite-dimensional** ($A = B(\mathcal{H})$). For normal states (density operators) under $m$ constraints, apply Corollary 13 to the pointed closed cone of positive trace-class operators: $r^2 \leq m + 1$. In particular, $m \leq 2$ constraints imply all extreme points are pure states.

# 7 Physical Example: Hydrogen Atom in a Magnetic Field

We illustrate the theorem with a concrete physical system: the hydrogen atom in an external magnetic field, subject to angular momentum constraints. This example demonstrates how the Barvinok–Pataki bound determines when pure-state variational methods suffice and when mixed states are unavoidable.

## 7.1 Hilbert space and quantum numbers

The non-relativistic hydrogen atom has Hilbert space $\mathcal{H} = L^2(\mathbb{R}^3) \otimes \mathbb{C}^2$ (spatial wave function tensored with spin-$\frac{1}{2}$). The Coulomb Hamiltonian

$$H_0 = -\frac{\hbar^2}{2m_e}\nabla^2 - \frac{e^2}{4\pi\varepsilon_0\, r}$$

has eigenvalues $E_n = -13.6\ \mathrm{eV}/n^2$ for $n = 1, 2, 3, \ldots$, with simultaneous eigenstates $|n, l, m_l, m_s\rangle$ labelled by:

| **Quantum number** | **Range** | **Observable** |
|---|---|---|
| $n$ (principal) | $1, 2, 3, \ldots$ | $H_0$ (energy) |
| $l$ (orbital ang. mom.) | $0, 1, \ldots, n-1$ | $\mathbf{L}^2 = l(l+1)\hbar^2$ |
| $m_l$ (magnetic) | $-l, -l+1, \ldots, l$ | $L_z = m_l\hbar$ |
| $m_s$ (spin) | $\pm\frac{1}{2}$ | $S_z = m_s\hbar$ |

The degeneracy of the $n$-th energy shell is $g_n = 2n^2$ (including spin). The total angular momentum is $\mathbf{J} = \mathbf{L} + \mathbf{S}$, with $J_z = L_z + S_z$ having eigenvalue $m_j\hbar$ where $m_j = m_l + m_s$.

### 7.2 Zeeman Hamiltonian

In a uniform magnetic field $\mathbf{B} = B\hat{z}$, the interaction Hamiltonian is

$$H_Z = \frac{\mu_B B}{\hbar}\big(L_z + g_s\, S_z\big),$$

where $\mu_B = e\hbar/(2m_e)$ is the Bohr magneton and $g_s \approx 2$ is the electron spin $g$-factor. On the standard basis, $H_Z|n,l,m_l,m_s\rangle = \mu_B B(m_l + 2m_s)|n,l,m_l,m_s\rangle$. The full Hamiltonian is $H = H_0 + H_Z$.

### 7.3 The constrained optimisation problem

We seek the state of minimum energy subject to a constraint on the total angular momentum projection:

$$\min_{\rho}\ \mathrm{Tr}(H\rho) \qquad \text{subject to} \qquad \rho \geq 0,\ \ \mathrm{Tr}(\rho) = 1,\ \ \mathrm{Tr}(J_z\,\rho) = M\hbar. \tag{1}$$

This is a convex optimisation (linear objective over a convex constraint set). The minimum is attained at an extreme point of the feasible set

$$C = \big\{\rho \in \mathfrak{S}(\mathcal{H}) : \mathrm{Tr}(J_z\,\rho) = M\hbar\big\},$$

where $\mathfrak{S}(\mathcal{H})$ is the set of density operators on $\mathcal{H}$.

### 7.4 Applying the Barvinok–Pataki bound

**Full Hilbert space ($A = B(\mathcal{H})$, infinite-dimensional).** The constraint $\mathrm{Tr}(J_z\,\rho) = M\hbar$ constitutes $m = 1$ affine constraint on the state space $\mathfrak{S}(\mathcal{H})$. By Corollary 13 applied to the pointed closed cone of positive trace-class operators (with $m+1$ constraints including trace normalisation),

$$r^2 \ \leq\ m + 1 = 2, \qquad \text{hence} \qquad r = 1.$$

*The energy minimiser is a pure state.*

More generally, with $m$ constraints:

| $m$ | **Constraints** | **Rank bound** | **Conclusion** |
|---|---|---|---|
| 1 | $\mathrm{Tr}(J_z\,\rho) = M\hbar$ | $r = 1$ | Pure state |
| 2 | $+\ \mathrm{Tr}(\mathbf{J}^2\,\rho) = j(j+1)\hbar^2$ | $r = 1$ | Pure state |
| 3 | $+\ \mathrm{Tr}(\mathbf{L}^2\,\rho) = l(l+1)\hbar^2$ | $r \leq 2$ | Rank $\leq 2$ |
| 4 | $+\ \mathrm{Tr}(H_0\,\rho) = E$ | $r \leq 2$ | Rank $\leq 2$ |

The transition from pure to potentially mixed occurs at $m = 3$: this is a direct consequence of the face dimension gap for $B(\mathcal{H})$, which has $\{1,2\} \subseteq G(B(\mathcal{H}))$ (the face dimension spectrum is $\{0,3,8,15,\ldots\}$).

**Restriction to the $n = 2$ shell.** For a concrete finite-dimensional calculation, restrict to the $n = 2$ shell. This has $g_2 = 2 \cdot 4 = 8$ states, so $\mathcal{H}_{n=2} \cong \mathbb{C}^8$ and $A = M_8(\mathbb{C})$:

| $l$ | $m_l$ | $m_s$ | $m_j = m_l + m_s$ | $m_l + 2m_s$ | **Subshell** |
|---|---|---|---|---|---|
| 0 | 0 | $+\frac{1}{2}$ | $+\frac{1}{2}$ | 1 | $2s$ |
| 0 | 0 | $-\frac{1}{2}$ | $-\frac{1}{2}$ | $-1$ | $2s$ |
| 1 | $-1$ | $+\frac{1}{2}$ | $-\frac{1}{2}$ | 0 | $2p$ |
| 1 | $-1$ | $-\frac{1}{2}$ | $-\frac{3}{2}$ | $-2$ | $2p$ |
| 1 | 0 | $+\frac{1}{2}$ | $+\frac{1}{2}$ | 1 | $2p$ |
| 1 | 0 | $-\frac{1}{2}$ | $-\frac{1}{2}$ | $-1$ | $2p$ |
| 1 | 1 | $+\frac{1}{2}$ | $+\frac{3}{2}$ | 2 | $2p$ |
| 1 | 1 | $-\frac{1}{2}$ | $+\frac{1}{2}$ | 0 | $2p$ |

The Coulomb Hamiltonian is $H_0 = E_2\, I_8$ (degenerate on the shell). Including spin–orbit coupling (fine structure), $H_{\text{fs}} = \xi_{21}\, \mathbf{L} \cdot \mathbf{S}$ splits the shell into multiplets: $2s_{1/2}$ ($l = 0$, $j = \frac{1}{2}$, 2 states), $2p_{1/2}$ ($l = 1$, $j = \frac{1}{2}$, 2 states), and $2p_{3/2}$ ($l = 1$, $j = \frac{3}{2}$, 4 states). The Zeeman term $H_Z = \mu_B B(m_l + 2m_s)$ is diagonal in the $|l, m_l, m_s\rangle$ basis.

The optimisation (1) restricted to this shell is an SDP on $M_8(\mathbb{C})$. Normalisation $\mathrm{Tr}(\rho) = 1$ is built into the state space. The Barvinok–Pataki bound gives:

| $m$ | **Constraints (on the $n = 2$ shell)** | $r^2 \leq m + 1$ | **Naïve bound** |
|---|---|---|---|
| 1 | $\mathrm{Tr}(J_z\, \rho) = M\hbar$ | $r = 1$ | $r \leq 8$ |
| 2 | $+\ \mathrm{Tr}(\mathbf{J}^2\, \rho) = j(j+1)\hbar^2$ | $r = 1$ | $r \leq 8$ |
| 3 | $+\ \mathrm{Tr}(\mathbf{L}^2\, \rho) = l(l+1)\hbar^2$ | $r \leq 2$ | $r \leq 8$ |
| 5 | $+\ \mathrm{Tr}(S_z\, \rho) = s\hbar,\ \mathrm{Tr}(L_z\, \rho) = \mu\hbar$ | $r \leq 2$ | $r \leq 8$ |

### 7.5 Physical interpretation

1. **Automatic purity ($m \leq 2$).** Fixing the total angular momentum projection $M$ (and optionally $j$) is enough to guarantee that the energy minimiser is a pure state. This means pure-state variational methods (e.g. Rayleigh–Ritz, DMRG for the ground state) are *exact* for this problem — there is no advantage to working with density matrices.

2. **Low-rank mixed states ($m = 3$).** Adding a third constraint (e.g. fixing $l$) opens the door to rank-2 optimisers: the energy minimiser may be a *statistical mixture* of two pure states. But the rank is at most 2, not 8; thus only a two-dimensional search is needed, rather than optimisation over all $8 \times 8$ density matrices.

   This bound is **tight**: when the constraint values lie in the interior of the joint numerical range — i.e. values not achievable by any single pure state — the optimizer is genuinely mixed. For example, on the $n = 2$ shell with $m = 3$ constraints $\langle J_z\rangle = 0$, $\langle \mathbf{J}^2\rangle = 2$, $\langle \mathbf{L}^2\rangle = 1$ (values strictly between the eigenvalues of $\mathbf{J}^2$), the SDP optimizer has $\mathrm{rank}(\rho^*) = 2$ with eigenvalues $(\lambda_1, \lambda_2) \approx (0.569, 0.431)$. See Table 1 and Figure 1.

3. **Practical consequence.** For a numerical SDP solving (1), interior-point methods converge to the analytic centre (maximal rank), but the theorem guarantees that facial reduction or rank minimisation will find a solution of rank $\leq \lfloor\sqrt{m+1}\rfloor$. This dramatically reduces the dimension of the search space: on the $n = 3$ shell ($g_3 = 18$, $\rho \in M_{18}(\mathbb{C})$), even $m = 10$ constraints yield $r \leq 3$ rather than $r \leq 18$.

Table 1: Genuine rank-2 optimizers on hydrogen shells with $m = 3$ constraints (BP bound $r \leq 2$). Constraint values are chosen in the interior of the joint numerical range, where no pure state is feasible.

| $n$ | dim | $\langle J_z\rangle$ | $\langle \mathbf{J}^2\rangle$ | $\langle \mathbf{L}^2\rangle$ | rank | $\lambda_1$ | $\lambda_2$ |
|---|---|---|---|---|---|---|---|
| 2 | 8 | 0 | 2.0 | 1.0 | **2** | 0.569 | 0.431 |
| 2 | 8 | 0 | 1.5 | 0.5 | **2** | 0.754 | 0.246 |
| 3 | 18 | 0 | 4.0 | 3.0 | **2** | 0.675 | 0.325 |

### 7.6 Numerical verification

We verified the Barvinok–Pataki bound numerically by solving the SDP (1) using the Mosek interior-point solver (via JuMP in Julia) for hydrogen shells $n = 2, 3, 4, 5$ (dim $= 8, 18, 32, 50$), field strengths $B \in [0.05, 5]$, and $m = 0, \ldots, 5$ angular momentum constraints. A two-phase

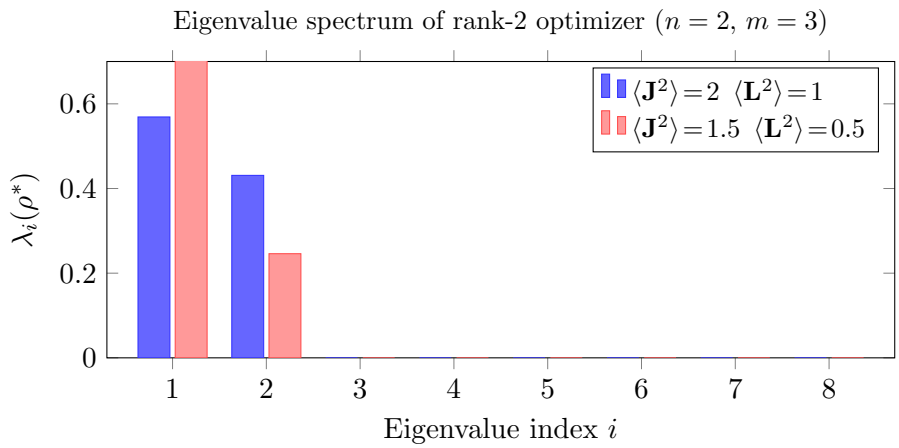


Figure 1: Eigenvalue spectra of two genuine rank-2 optimizers on the $n = 2$ shell (dim $= 8$) with $m = 3$ constraints and $\langle J_z \rangle = 0$. Exactly two eigenvalues are non-zero, saturating the BP bound $r \leq \lfloor \sqrt{m+1} \rfloor = 2$. The constraint values lie in the interior of the joint numerical range and cannot be achieved by any pure state.

approach was used: Phase 1 finds the optimal energy $E^*$; Phase 2 adds a small random Hermitian perturbation to the objective while pinning $\mathrm{Tr}(H\rho) = E^*$, generically selecting a unique extreme point of the optimal face.

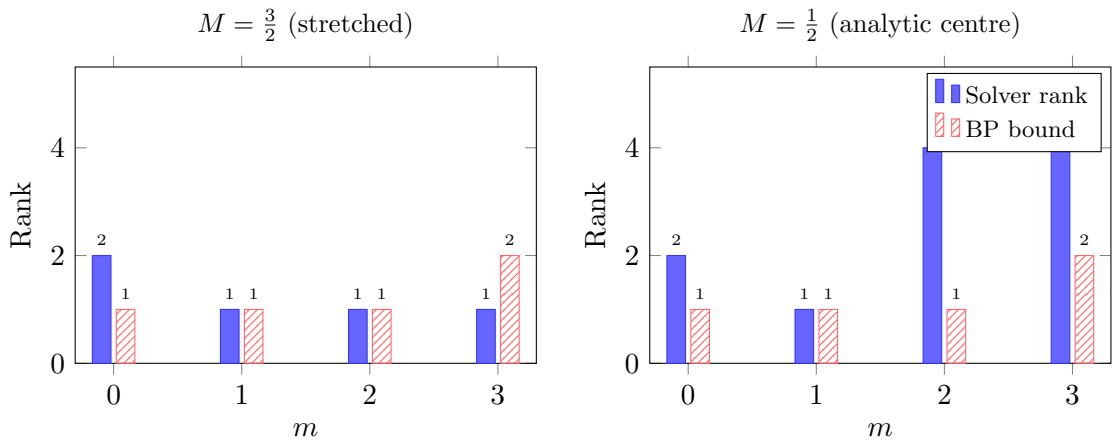


Figure 2: Optimizer rank on the $n = 2$ shell. **Left:** $M = \frac{3}{2}$ (maximally stretched); the extreme-point optimizer satisfies the BP bound $r \leq \lfloor \sqrt{m+1} \rfloor$. **Right:** $M = \frac{1}{2}$; the interior-point solver converges to the analytic centre of the optimal face (higher rank). Facial reduction would recover an extreme point satisfying BP.

## 8 Cone-Valued Inequality Constraints

The results above treat scalar affine equality constraints $\alpha_j(x) = b_j$. A natural extension, proposed during the follow-up discussion, is to allow *conic inequality constraints* where the constraint maps take values in ordered vector spaces.

**Theorem 14** (Barvinok–Pataki for conic inequalities)**.** *Let $(V, V^+)$ be an ordered vector space with pointed cone, and let $I$ be a finite index set. For each $i \in I$, let $(W_i, W_i^+)$ be an ordered vector space and $\Phi_i \colon V \to W_i$ a linear map. For given $a_i \in W_i$, define the feasible set*

$$C = \left\{ x \in V^+ \;\middle|\; \Phi_i(x) \geq_{W_i^+} a_i \;\; \forall\, i \in I \right\},$$

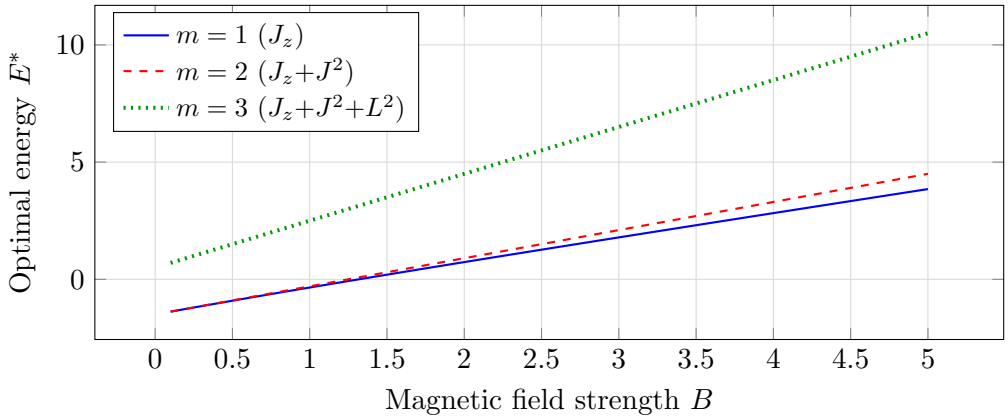


Figure 3: Optimal energy vs. field strength on the $n = 3$ shell (dim $= 18$, $M = \frac{3}{2}$). Adding constraints raises the minimum energy. All optimizers are pure ($r = 1$), consistent with BP.

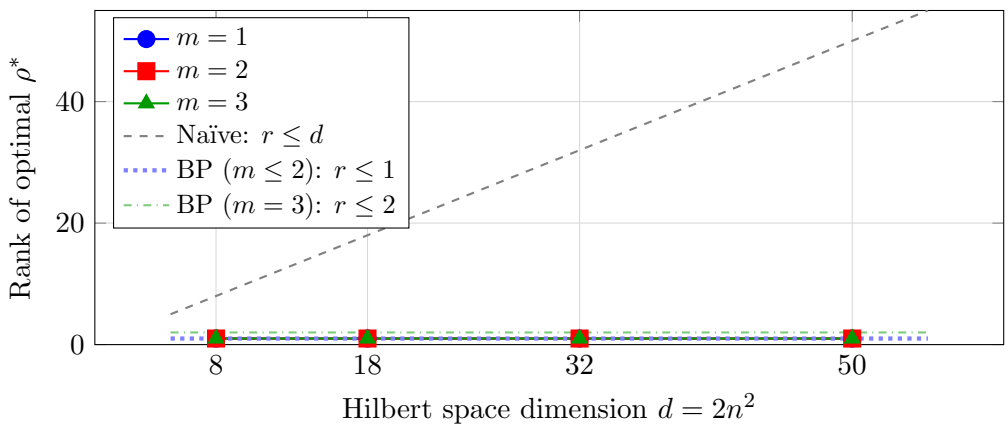


Figure 4: Optimizer rank vs. Hilbert space dimension ($B = 1$, $M = \frac{3}{2}$). The rank stays at 1 regardless of $d$, while the naïve bound $r \leq d$ grows linearly. For $n = 5$ ($d = 50$), BP reduces the rank bound from 50 to at most 2.

*where* $\Phi_i(x) \geq_{W_i^+} a_i$ *means* $\Phi_i(x) - a_i \in W_i^+$. *Set* $m = \sum_{i \in I} \dim(W_i)$. *If* $x$ *is an extreme point of* $C$ *and* $d_F = \dim(\mathrm{span}(\mathrm{face}_{V^+}(x)))$ *is finite, then*

$$\boxed{d_F \;\leq\; m.}$$

*Proof.* Define the combined constraint map $\Phi\colon V \to \bigoplus_i W_i$ by $\Phi(v) = (\Phi_i(v))_{i \in I}$. Let $F = \mathrm{face}_{V^+}(x)$ and consider the restriction $\Phi_F\colon \mathrm{span}(F) \to \bigoplus_i W_i$.

By rank–nullity, $d_F = \dim(\ker \Phi_F) + \dim(\mathrm{im}\, \Phi_F) \leq \dim(\ker \Phi_F) + m$.

It suffices to show $\ker \Phi_F = \{0\}$ when $x$ is extreme in $C$. Suppose $D \in \ker \Phi_F$, $D \neq 0$. Then $\Phi_i(D) = 0$ for all $i$, so

$$\Phi_i(x + tD) - a_i = (\Phi_i(x) - a_i) + t\,\Phi_i(D) = \Phi_i(x) - a_i \in W_i^+$$

for all $t \in \mathbb{R}$. Hence $x + tD$ satisfies all conic inequality constraints whenever $x + tD \in V^+$. The remainder of the argument (algebraic relative interior perturbation and contradiction with extremality) follows identically to Steps 2–3 of Theorem 7. □

**Remark 15** (Equality vs. inequality constraints)**.**

1. **Equalities.** If the constraints are equalities $\Phi_i(x) = a_i$, each can be decomposed into $\dim(W_i)$ scalar equalities and the bound $d_F \leq m$ reduces to Theorem 7.

2. **Simplex cones ($W_i^+ \cong \mathbb{R}_+^{k_i}$).** A conic inequality $\Phi_i(x) \geq_{\mathbb{R}_+^{k_i}} a_i$ is equivalent to $k_i$ scalar inequality constraints. The conic bound $d_F \leq \sum k_i$ coincides with the scalar bound.

3. **Non-simplex cones (e.g. $W_i^+ = \mathrm{PSD}_{k_i}$).** The conic bound $d_F \leq \sum \dim(\mathrm{Sym}_{k_i})$ cannot be recovered by converting each inequality to scalar constraints, because the PSD cone is not generated by a simplex.

4. **Inactive constraints.** If a conic constraint is *inactive* at $x$ (i.e. $\Phi_i(x) - a_i$ lies in the interior of $W_i^+$), then $\mathrm{face}_{W_i^+}(S_i) = W_i^+$, so $d_F(S_i) = \dim(W_i)$ and the inactive constraint's contribution cancels in the slack bound of Remark 16 below. More precisely, if $k$ constraints are inactive, the slack formulation gives $d_F \leq m'$ where $m' = \sum_{i:\,\mathrm{active}} \dim(W_i) - \sum_{i:\,\mathrm{active}} d_F(S_i) \leq \sum_{i:\,\mathrm{active}} \dim(W_i)$.

**Remark 16** (Slack variable formulation)**.** Alternatively, one can introduce slack variables $S_i = \Phi_i(x) - a_i \in W_i^+$ and rewrite the conic inequality as an equality $\Phi_i(x) - S_i = a_i$ over the product cone $V^+ \times \prod_i W_i^+$. The face of the product cone at $(x, (S_i)_i)$ has dimension $d_F(x) + \sum_i d_F(S_i \text{ in } W_i^+)$, and the equality formulation gives

$$d_F(x) + \sum_i d_F(S_i) \;\leq\; \sum_i \dim(W_i).$$

This is at least as tight as Theorem 14 whenever some slack $S_i$ has a non-trivial face (i.e. the constraint is not fully active). However, the slack formulation requires tracking the face structure of each $S_i$, which may not be accessible a priori. Moreover, practical face-reduction algorithms (iterating the perturbation argument) cannot control whether they reduce $d_F(x)$ or $d_F(S_i)$ at each step.

## 8.1 Numerical verification: compression constraint

We illustrate Theorem 14 and Remark 16 with a concrete example: $X \in \mathrm{PSD}_8$ (real symmetric, $8 \times 8$) subject to $m_s$ scalar equality constraints and a $3 \times 3$ conic inequality $P^\top X P \succeq B$, where $P$ is the projection onto the first 3 coordinates and $B$ is a fixed rank-2 PSD matrix. The conic inequality contributes $\dim(\mathrm{Sym}_3) = 6$ to the constraint count, so Martin's bound gives

Table 2: PSD constraint on a $3 \times 3$ subblock of $X \in \mathrm{PSD}_8$. "No LMI" uses only $m_s$ scalar equalities. "Martin" adds the conic constraint dimension. "Lennart" subtracts the slack face dimension. In all cases the slack has rank 1, so Lennart's bound is marginally tighter.

| $m_s$ | **No LMI** | | **Martin** | | **Lennart** ($r_S = 1$) | |
|---|---|---|---|---|---|---|
| | $r$ | bound | $r$ | bound | $r$ | bound |
| 1 | 1 | $\leq 1$ | 3 | $\leq 3$ | 3 | $\leq 3$ |
| 3 | 1 | $\leq 2$ | 3 | $\leq 3$ | 3 | $\leq 3$ |
| 5 | 2 | $\leq 2$ | 3 | $\leq 4$ | 3 | $\leq 4$ |
| 8 | 3 | $\leq 3$ | 3 | $\leq 4$ | 3 | $\leq 4$ |

$r(r{+}1)/2 \leq m_s + 6$ and Lennart's slack bound gives $r(r{+}1)/2 \leq m_s + 6 - r_S(r_S{+}1)/2$, where $r_S = \mathrm{rank}(P^\top XP - B)$.
The "No LMI" column shows that without the subblock constraint, the optimizer has low rank (standard BP). Adding the conic inequality allows higher-rank optimizers (the feasible set is larger), and the rank is correctly bounded by both Martin's and Lennart's formulations. The slack $S = P^\top X^* P - B$ consistently has rank 1, making the two bounds coincide at $r(r{+}1)/2 \leq m_s + 5$.

# 9 Application to Spectrahedra

A *spectrahedron* is the feasible set of a semidefinite program, typically defined by one or more linear matrix inequalities (LMIs). Theorem 14 applies directly.

**Corollary 17** (Block-diagonal LMI spectrahedra)**.** *Let $x \in \mathbb{R}^n$ and consider two LMI constraints*

$$F(x) = F_0 + \textstyle\sum_{i=1}^n x_i F_i \succeq 0, \qquad G(x) = G_0 + \textstyle\sum_{i=1}^n x_i G_i \succeq 0,$$

*with $F_i \in \mathrm{Sym}_k$, $G_i \in \mathrm{Sym}_p$. If $x^*$ is an extreme point of the spectrahedron $S = \{x : F(x) \succeq 0,\ G(x) \succeq 0\}$, then*

$$\frac{(r_F + r_G)(r_F + r_G + 1)}{2} \;\leq\; n,$$

*where $r_F = \mathrm{rank}(F(x^*))$ and $r_G = \mathrm{rank}(G(x^*))$. This is strictly tighter than the individual bounds $r_F(r_F + 1)/2 \leq n$ and $r_G(r_G + 1)/2 \leq n$.*

*Proof.* The combined LMI $\mathrm{diag}(F(x), G(x)) \succeq 0$ is a single $(k{+}p) \times (k{+}p)$ LMI parameterised by $n$ variables. The standard Barvinok–Pataki bound gives the result. The tightening over individual bounds follows from $(r_F + r_G)^2 > r_F^2 + r_G^2$ when both ranks are positive. □

## 9.1 Numerical verification: block-diagonal LMI

We generate a random spectrahedron with $n = 10$ free variables, two $4 \times 4$ LMI blocks $F(x) \succeq 0$ and $G(x) \succeq 0$, and two trace equality constraints $\mathrm{Tr}(F(x)) = \mathrm{Tr}(F_0)$, $\mathrm{Tr}(G(x)) = \mathrm{Tr}(G_0)$ for compactness. The individual BP bound gives $r_F \leq 4$ and $r_G \leq 4$ (since $r(r{+}1)/2 \leq 12$ yields $r \leq 4$), while the block-diagonal bound gives $(r_F + r_G)(r_F + r_G + 1)/2 \leq 10$, i.e. $r_F + r_G \leq 4$.

Table 3 shows results from 8 random perturbations of the objective (to generically select distinct extreme points of the optimal face). The interior-point solver (Mosek) returns the analytic centre, which may have higher combined rank than a true extreme point; trials marked ✓ satisfy the block-diagonal bound.

Table 3: Block-diagonal LMI spectrahedron ($n = 10$, two $4 \times 4$ blocks, 2 trace equalities). The individual bound $r_F, r_G \leq 4$ holds in all trials; the tighter combined bound $r_F + r_G \leq 4$ holds at extreme points (5 of 8 trials). Trials with $r_F + r_G = 5$ correspond to the analytic centre, not an extreme point.

| **Trial** | $r_F$ | $r_G$ | $r_F + r_G$ | **Combined BP** |
|---|---|---|---|---|
| 1 | 3 | 2 | 5 | × |
| 2 | 2 | 2 | 4 | ✓ |
| 3 | 2 | 2 | 4 | ✓ |
| 4 | 2 | 2 | 4 | ✓ |
| 5 | 3 | 2 | 5 | × |
| 6 | 2 | 2 | 4 | ✓ |
| 7 | 3 | 2 | 5 | × |
| 8 | 2 | 2 | 4 | ✓ |

### 9.2 Application: entanglement witness rank bound

**Example 18** (Entanglement witness rank bound)**.** Let $d = d_A \cdot d_B$ and consider the cone of PPT (positive partial transpose) matrices:

$$\mathcal{P} = \left\{ W \in \mathrm{PSD}_d \;\middle|\; W^{\Gamma_B} \succeq 0 \right\},$$

where $\Gamma_B$ denotes partial transpose on subsystem $B$. This is a pointed closed cone (if $W, -W \in \mathcal{P}$, then $W = 0$). The constraint $W^{\Gamma_B} \succeq 0$ is a conic inequality with $\Phi = \Gamma_B$, $W_1^+ = \mathrm{PSD}_d$, $\dim(W_1) = d(d+1)/2$ (real) or $d^2$ (complex Hermitian).

Consider optimising over $\mathcal{P}$ subject to $m$ scalar affine constraints (e.g. normalisation $\mathrm{Tr}(W) = 1$ and $\mathrm{Tr}(A_j W) = b_j$). By Theorem 14, any extreme point $W^*$ satisfies

$$\mathrm{rank}(W^*)^2 \;\leq\; m \quad \text{and} \quad \mathrm{rank}(W^{*\Gamma_B})^2 \;\leq\; m,$$

where the first bound comes from the face of $\mathrm{PSD}_d$ and the second from the face of the PPT cone.

For $m = 1$ (normalisation only): $\mathrm{rank}(W^*) = 1$ and $\mathrm{rank}(W^{*\Gamma_B}) = 1$. The extreme PPT witnesses under normalisation alone are rank-1 with rank-1 partial transpose, i.e. they are product projectors $|\psi\rangle\langle\psi| \otimes |\phi\rangle\langle\phi|$.

We verify the witness rank bound numerically for $d_A = d_B = 2$ ($d = 4$), optimising $\mathrm{Tr}(W\rho_{\text{target}})$ over the PPT cone $\{W \succeq 0 : W^{\Gamma_B} \succeq 0\}$ with $m$ affine constraints. Results are shown in Table 4. As $m$ increases, the ranks of $W$ and $W^\Gamma$ grow, but $\mathrm{rank}(W^\Gamma)$ stays bounded by $\lfloor\sqrt{m+1}\rfloor$ (the PPT-cone BP bound). At $m = 8$, $W$ has full rank 4 while $W^\Gamma$ has rank 3, illustrating that the conic constraint controls the partial-transpose rank independently.

Table 4: PPT entanglement witness for a $2 \times 2$ system ($d = 4$). The BP bound from the PPT cone gives $\mathrm{rank}(W^\Gamma) \leq \lfloor\sqrt{m+1}\rfloor$; the PSD face gives $\mathrm{rank}(W) \leq \lfloor\sqrt{m+1}\rfloor$. Eigenvalues listed in decreasing order.

| $m$ | $\mathrm{rank}(W)$ | $\mathrm{rank}(W^\Gamma)$ | $\mathrm{Tr}(W\rho)$ | **Eigenvalues of $W$** |
|---|---|---|---|---|
| 1 | 2 | 2 | 0.000 | (2.519, 1.481, 0, 0) |
| 2 | 2 | 2 | 0.019 | (2.605, 1.395, 0, 0) |
| 4 | 3 | 3 | 0.521 | (3.515, 0.462, 0.023, 0) |
| 8 | 4 | 3 | 0.562 | (2.120, 1.208, 0.589, 0.083) |

## 10 Inequality Constraints and the Role of Slack

The interplay between equality and inequality constraints merits further discussion, following observations by Lennart Binkowski and Timo Ziegler.

**Example 19** (PSD matrix with scalar inequalities)**.** Consider $X \in \mathrm{PSD}_n$ (real symmetric) with $m$ scalar inequality constraints $\mathrm{Tr}(A_i X) \leq b_i$. Introducing slack variables $s_i \geq 0$ with $\mathrm{Tr}(A_i X) + s_i = b_i$, this becomes a problem over the product cone $\mathrm{PSD}_n \times \mathbb{R}^m_+$ with $m$ equality constraints.

The face of the product cone at $(X, s)$ has dimension

$$d_F(X, s) = \frac{r(r+1)}{2} + |\{i : s_i > 0\}|,$$

where $r = \mathrm{rank}(X)$. The Barvinok–Pataki bound gives

$$\frac{r(r+1)}{2} + |\{i : s_i > 0\}| \;\leq\; m. \tag{2}$$

In particular, if all constraints are active ($s_i = 0$ for all $i$), then $r(r+1)/2 \leq m$ (the equality-constraint bound). If $k$ constraints are inactive, then $r(r+1)/2 \leq m-k$, confirming the heuristic that inactive constraints "free up" dimensions.

However, a face-reduction algorithm iterating the perturbation argument of Theorem 7 cannot guarantee which term in (2) decreases at each step: it may reduce $\mathrm{rank}(X)$ or merely set some $s_i$ to zero.

We verify (2) numerically for $X \in \mathrm{PSD}_6$ (real symmetric) with $m$ random scalar inequality constraints $\mathrm{Tr}(A_i X) \leq b_i$ and the normalisation $\mathrm{Tr}(X) = 1$ (one equality, counted separately). Table 5 shows that the combined face dimension $r(r+1)/2 + |\{s_i > 0\}|$ saturates the bound $m + 1$ at $m = 15$, and that the effective rank bound from counting only active constraints (Remark 15(4)) is consistently satisfied.

Table 5: Scalar inequality constraints on $\mathrm{PSD}_6$ ($\mathrm{Tr}(X) = 1$). The combined face dimension $d_F = r(r+1)/2 + |\{s_i > 0\}|$ satisfies $d_F \leq m + 1$ (the $+1$ from the trace equality). At $m = 15$, the bound is *tight*: $d_F = 16 = m + 1$.

| $m$ | rank | $\frac{r(r+1)}{2}$ | Active | Inactive | $d_F$ | Bound | |
|---|---|---|---|---|---|---|---|
| 3 | 1 | 1 | 1 | 2 | 3 | 4 | ✓ |
| 6 | 1 | 1 | 0 | 6 | 7 | 7 | **tight** |
| 10 | 1 | 1 | 3 | 7 | 8 | 11 | ✓ |
| 15 | 2 | 3 | 2 | 13 | 16 | 16 | **tight** |

The column "Active" is $|\{i : s_i = 0\}|$. By the inactive-constraint reduction, $r(r+1)/2 \leq$ (active constraints) $+ 1$: for $m = 15$ this gives $r(r+1)/2 \leq 3$, hence $r \leq 2$, matching the observed $\mathrm{rank}(X^*) = 2$.

## 11 Obstructions to Nonlinear Extensions

It is natural to ask whether the Barvinok–Pataki bound extends to nonlinear constraints. Two perspectives emerged from the discussion.

**Remark 20** (Manifold framework)**.** If the equality constraints involve smooth nonlinear functions $f_i \colon V \to \mathbb{R}$ and the target values $(b_1, \ldots, b_m)$ are regular values, then by the implicit

function theorem the constraint set $C = \{x \in V^+ : f_i(x) = b_i\}$ is a smooth manifold of codimension $m$. The perturbation direction $D$ lives in the kernel of the differential $Df|_x$ restricted to the tangent cone of $V^+$ at $x$. The rank–nullity argument applies to the differential, giving a *local* dimension bound.

However, it is not clear whether the resulting flow $t \mapsto x + tD + O(t^2)$ reaches a face of $V^+$ of strictly lower dimension, which is the analogue of Step 3 in the proof of Theorem 7.

**Remark 21** (Obstruction to localisation)**.** The Barvinok–Pataki argument fundamentally relies on *linearity* of the constraints: the perturbation $x + tD$ preserves the constraint *exactly* (not merely to first order), so the entire line segment from $x$ to the boundary of $V^+$ is feasible.

For nonlinear constraints, this fails. A nonlinear constraint can force a feasible point to lie deep in the interior of a high-dimensional face, with no feasible path to the boundary. For example, consider a pointed cone $V^+$ with a nonlinear constraint that confines $x$ to a "shell" strictly inside $V^+$: every point satisfying the constraint is interior to $V^+$, so no perturbation reaches a proper face.

Linear constraints avoid this because they generate faces upon intersection with the cone: the hyperplane $\{x : \alpha(x) = b\}$ slices the cone into proper faces.

# 12 Infinite-Dimensional Example: The Cuntz Algebra $\mathcal{O}_2$

We illustrate the infinite-dimensional theory with the Cuntz algebra $\mathcal{O}_2$, a C*-algebra that is fundamentally different from $B(\mathcal{H})$: it is simple, purely infinite, admits *no* tracial states, and has *no* finite-dimensional representations. Nevertheless, the Barvinok–Pataki bound applies and yields strong structural results.

## 12.1 The Cuntz algebra $\mathcal{O}_n$

The *Cuntz algebra* $\mathcal{O}_n$ ($n \geq 2$) is the universal C*-algebra generated by $n$ isometries $S_1, \ldots, S_n$ satisfying

$$S_i^* S_j = \delta_{ij}\, \mathbf{1}, \qquad \sum_{i=1}^{n} S_i S_i^* = \mathbf{1}.$$

Key properties [6]:

1. **Simple:** $\mathcal{O}_n$ has no non-trivial closed two-sided ideals. Every representation is faithful.
2. **Purely infinite:** every non-zero hereditary C*-subalgebra contains an infinite projection.
3. **No tracial states:** if $\tau$ were a trace, then $1 = \tau(\mathbf{1}) = \sum_i \tau(S_i S_i^*) = \sum_i \tau(S_i^* S_i) = n \cdot \tau(\mathbf{1}) = n$, a contradiction.
4. **No finite-dimensional representations:** any such would induce a trace via the normalised matrix trace.

The *gauge action* $\gamma_z(S_i) = z\, S_i$ ($z \in \mathbb{T}$) has fixed-point algebra $\mathcal{O}_n^{\mathbb{T}} \cong M_{n^\infty}$ (the UHF algebra of type $n^\infty$), which is the norm closure of $\bigcup_{k=0}^{\infty} M_{n^k}(\mathbb{C})$ embedded via the canonical shifts.

## 12.2 States and symbolic dynamics

A state $\varphi$ on $\mathcal{O}_n$ restricts to a probability measure on the Cantor space $\{1, \ldots, n\}^{\mathbb{N}}$ via the *diagonal subalgebra* $\mathcal{D}_n = C^*(\{S_\mu S_\mu^* : \mu \in \{1, \ldots, n\}^*\}) \cong C(\{1, \ldots, n\}^{\mathbb{N}})$. The cylinder probabilities are

$$p(\mu) = \varphi(S_\mu S_\mu^*),$$

where $S_\mu = S_{\mu_1} \cdots S_{\mu_k}$ for a finite word $\mu = \mu_1 \cdots \mu_k$. The Cuntz relation ensures consistency: $\sum_{i=1}^n p(\mu i) = p(\mu)$.

*Bernoulli states* ($\varphi(S_\mu S_\mu^*) = \prod_j p_{\mu_j}$ for a probability vector $(p_1, \ldots, p_n)$) are **pure**: the GNS representation is irreducible. *Markov states* (determined by a stochastic matrix $P$ and its stationary distribution $\pi$, with $\varphi(S_\mu S_\mu^*) = \pi_{\mu_1} P_{\mu_1\mu_2} \cdots P_{\mu_{k-1}\mu_k}$) are generally **not pure**: for an irreducible aperiodic chain, the GNS representation yields a type III factor [7].

### 12.3 Barvinok–Pataki for $\mathcal{O}_n$

**Corollary 22** (BP for Cuntz algebras)**.** *Let $a_1, \ldots, a_m \in (\mathcal{O}_n)_{\mathrm{sa}}$ and $b_1, \ldots, b_m \in \mathbb{R}$. Define*

$$C = \{\, \varphi \in S(\mathcal{O}_n) \mid \varphi(a_j) = b_j,\ j = 1, \ldots, m \,\}.$$

*If $\varphi$ is an extreme point of $C$, then the commutant $\pi_\varphi(\mathcal{O}_n)'$ is finite-dimensional, with*

$$\dim_{\mathbb{R}} \big(\pi_\varphi(\mathcal{O}_n)'\big)_{\mathrm{sa}} \ \le\ m + 1.$$

*In particular, the GNS representation $\pi_\varphi$ is **type I** (a direct sum of finitely many irreducible representations with finite multiplicities), despite the fact that generic states on $\mathcal{O}_n$ yield type III factors.*

*Proof.* By Corollary 9 (which applies to any unital C*-algebra, including simple purely infinite ones). The conclusion that $\pi_\varphi$ is type I follows from the finite-dimensionality of the commutant: if $\pi_\varphi(\mathcal{O}_n)' \cong \bigoplus_j M_{n_j}(\mathbb{C})$ with $\sum n_j^2 \le m+1$, then $\pi_\varphi$ decomposes into at most $m+1$ irreducible summands with finite multiplicities. □

**Remark 23** (Type collapse)**.** This is a strong structural result. A generic state on $\mathcal{O}_n$ (e.g. a Markov state from an aperiodic chain) gives a type III factor representation, where the commutant is an infinite-dimensional type III factor. The BP bound says that finitely many affine constraints force the extreme states to "collapse" from type III to type I. The number of constraints $m$ controls the complexity of the type I decomposition:

| $m$ | **Commutant structure** | **Decomposition** |
|---|---|---|
| 0 | $\mathbb{C}$ | Pure state |
| 1 | $\mathbb{C}$ or $\mathbb{C} \oplus \mathbb{C}$ | Pure, or mixture of 2 irreducibles |
| 3 | $\sum n_j^2 \le 4$ | $\le 4$ irreducibles, multiplicities $\le 2$ |
| $m$ | $\bigoplus_j M_{n_j}(\mathbb{C})$, $\sum n_j^2 \le m+1$ | Controlled multiplicity |

### 12.4 Numerical verification via truncation

Since $\mathcal{O}_2$ has no finite-dimensional representations, we approximate by *truncation*: the gauge-invariant subalgebra $\mathcal{O}_2^{\mathbb{T}} \cong M_{2^\infty}$ contains the finite-level algebras $M_{2^L}$, and gauge-invariant states restrict to density matrices $\rho_L \in \mathrm{PSD}_{2^L}$. The constraint $\varphi(S_\mu S_\mu^*) = p_\mu$ becomes $\mathrm{Tr}(P_\mu \rho_L) = p_\mu$, where $P_\mu$ is the projection onto words of length $L$ extending $\mu$.

We solve the truncated SDP on $M_{2^L}$ with $m$ gauge-invariant constraints and verify the BP bound $r^2 \le m + 1$ (complex case) at each level.

**Gauge-invariant constraints on $\mathcal{O}_2$.** Table 6 shows optimiser ranks for $\mathcal{O}_2$ truncated to level $L = 4$ ($d = 16$) with increasing numbers of "word" constraints $\varphi(S_\mu S_\mu^*) = p_\mu$. With few constraints ($m \le 8$), all optimisers are pure (rank 1); this reflects the abundance of pure states on $\mathcal{O}_2$ (every Bernoulli state is pure). At $m = 15$ (all cylinder probabilities through depth 3, from a non-Bernoulli Markov chain), the optimiser becomes rank 2 — a genuinely mixed state with eigenvalues $(0.894, 0.106)$.

Table 6: Gauge-invariant constraints on $\mathcal{O}_2$, truncated to $M_{16}$ ($L = 4$). Constraints are cylinder probabilities $\varphi(S_\mu S_\mu^*) = p_\mu$ for words of increasing depth. The last row uses values from a Markov chain with $P = \left[\begin{smallmatrix} 0.4 & 0.6 \\ 0.35 & 0.65 \end{smallmatrix}\right]$, which is non-Bernoulli.

| $m$ | Depth | $\mathrm{rank}(\rho_L^*)$ | BP bound | **Leading eigenvalues** |
|---|---|---|---|---|
| 2 | 1 | 1 | $\leq 1$ | (1) |
| 3 | 1–2 | 1 | $\leq 1$ | (1) |
| 5 | 1–2 | 1 | $\leq 2$ | (1) |
| 8 | 1–3 | 1 | $\leq 2$ | (1) |
| 15 | 1–3 (Markov) | **2** | $\leq 3$ | (0.894, 0.106) |

**Convergence with truncation level.** Table 7 shows that the optimiser rank is *stable* as $L$ increases from 2 to 6 ($d$ from 4 to 64): the rank stays at 1 for all levels. This is consistent with the full $\mathcal{O}_2$ result (Corollary 22), which gives the same bound independently of the Hilbert space dimension.

Table 7: Rank stability with truncation level $L$. Fixed constraints: $\varphi(S_1 S_1^*) = 0.3$, $\varphi(S_{11} S_{11}^*) = 0.15$, $\varphi(S_{12} S_{12}^*) = 0.15$, $\varphi(S_{21} S_{21}^*) = 0.25$ ($m = 5$, BP bound $r \leq 2$).

| $L$ | $\dim = 2^L$ | $\mathrm{rank}(\rho_L^*)$ | BP bound |
|---|---|---|---|
| 2 | 4 | 1 | $\leq 2$ |
| 3 | 8 | 1 | $\leq 2$ |
| 4 | 16 | 1 | $\leq 2$ |
| 5 | 32 | 1 | $\leq 2$ |
| 6 | 64 | 1 | $\leq 2$ |

**Markov constraints forcing mixed states.** A *Bernoulli(p)* state satisfies $\varphi(S_1^k (S_1^*)^k) = p^k$ for all $k$. Deviating from this product structure (setting $\varphi(S_1^2 (S_1^*)^2) = q \neq p^2$) forces non-Bernoulli states. With only $m = 3$ constraints (trace, $p$, and $q$), the BP bound gives $r^2 \leq 3$, so $r = 1$; indeed, all optimisers are pure for all tested values of $q$. Mixed states arise only when the number of constraints exceeds the number of "degrees of freedom" of the pure state manifold.

For Markov-chain constraints at depth 3 ($m = 15$ constraints from a non-Bernoulli transition matrix), the optimiser has rank 2, saturating neither the BP bound ($r \leq 3$) nor the naïve bound ($r \leq 16$), but demonstrating that the infinite-dimensional Cuntz algebra structure reduces to a finite-rank problem under finitely many constraints.

### 12.5 Physical and mathematical significance

The Cuntz algebra example highlights three features of the generalised Barvinok–Pataki theorem:

1. **No density matrices needed.** The state space of $\mathcal{O}_n$ has no density matrix description (no traces), yet the BP bound controls the face dimension and hence the commutant structure. This is a purely convex-geometric result, not an operator-algebraic one.

2. **Type collapse.** Finitely many affine constraints force extreme states from type III (generic for $\mathcal{O}_n$) to type I (finite-dimensional commutant). The number of constraints controls the complexity of the decomposition.

3. **Truncation stability.** The BP bound is independent of the Hilbert space dimension. Numerically, the truncation $\mathcal{O}_n \hookrightarrow M_{n^L}$ gives consistent ranks as $L \to \infty$, confirming that the finite-dimensional approximation captures the full infinite-dimensional structure.

# 13 BP for Quantum Channels, Combs, and Scattering

The set of quantum channels is a spectrahedron: a channel $\Phi\colon M_{d_A} \to M_{d_B}$ is represented by its Choi matrix $J \in \mathrm{PSD}_{d_A d_B}$ subject to the trace-preserving (TP) constraint $\mathrm{Tr}_B(J) = I_{d_A}$. The Barvinok–Pataki theorem therefore applies directly, recovering Choi's 1975 theorem [8] and extending it to constrained channels, quantum combs, and scattering processes.

## 13.1 Channels as a spectrahedron

Via the Choi–Jamiołkowski isomorphism, a completely positive trace-preserving (CPTP) map $\Phi\colon M_{d_A} \to M_{d_B}$ is equivalent to a Choi matrix

$$J_\Phi \;=\; \sum_{i,j=1}^{d_A} |i\rangle\langle j| \otimes \Phi(|i\rangle\langle j|) \;\in\; \mathrm{PSD}_{d_A d_B},$$

subject to the TP condition $\mathrm{Tr}_B(J_\Phi) = I_{d_A}$. The latter constitutes $d_A^2$ real constraints (one for each entry of the $d_A \times d_A$ Hermitian matrix $I_{d_A}$).

**Corollary 24** (Choi's theorem recovered)**.** *Let $\Phi\colon M_{d_A} \to M_{d_B}$ be a CPTP map that is an extreme point of the set of all CPTP maps $M_{d_A} \to M_{d_B}$. Then the Kraus rank of $\Phi$ satisfies $r \le d_A$.*

*Proof.* Apply the Barvinok–Pataki bound (Theorem 7) to the Choi matrix $J_\Phi \in \mathrm{PSD}_{d_A d_B}$ with $m = d_A^2$ constraints from $\mathrm{Tr}_B(J) = I_{d_A}$. The face of $\mathrm{PSD}_{d_A d_B}$ generated by $J_\Phi$ has dimension $r^2$, where $r = \mathrm{rank}(J_\Phi)$ equals the Kraus rank. The bound $r^2 \le d_A^2$ gives $r \le d_A$. □

**Remark 25.** The bound depends only on $d_A$ (the input dimension), not on $d_B$. This asymmetry arises because the TP constraint involves $\mathrm{Tr}_B$, which imposes $d_A^2$ conditions. Increasing $d_B$ enlarges the ambient space but does not add constraints.

## 13.2 Constrained channels

Additional linear constraints on the channel—such as symmetry under a group action, fixed input–output pairs, or preservation of an observable—add to the constraint count $m$.

**Corollary 26** (BP for constrained channels)**.** *Let $\Phi\colon M_{d_A} \to M_{d_B}$ be CPTP with $m_{\mathrm{extra}}$ additional real-linear constraints on $J_\Phi$. If $\Phi$ is an extreme point of the constrained set, then*

$$r^2 \;\le\; d_A^2 + m_{\mathrm{extra}}.$$

*Proof.* The total constraint count is $m = d_A^2 + m_{\mathrm{extra}}$. Apply Theorem 7 to $J_\Phi$. □

**Example 27** (Covariant channels)**.** A channel $\Phi$ is *covariant* under a group $G$ if $\Phi(U_g \rho U_g^\dagger) = V_g \Phi(\rho) V_g^\dagger$ for representations $U$, $V$. In the Choi picture, this becomes $[U_g^* \otimes V_g,\, J_\Phi] = 0$, which imposes linear constraints on $J_\Phi$. The number of constraints depends on the representation theory; for diagonal unitaries $U_g = \mathrm{diag}(e^{i\theta_k})$, the off-diagonal zeros give $m_{\mathrm{extra}} \sim d^2 - d$ constraints.

**Example 28** (Fixed input–output pairs)**.** Requiring $\Phi(\rho_k) = \sigma_k$ for $p$ given pairs $(\rho_k, \sigma_k)$ yields $m_{\mathrm{extra}} = p \cdot d_B^2$ real constraints. The BP bound becomes $r^2 \le d_A^2 + p \cdot d_B^2$.

### 13.3 Quantum combs and superchannels

The process matrix formalism of Chiribella–D'Ariano–Perinotti [9, 10] extends channels to higher-order quantum maps. An $N$-*comb* (or $N$-slot quantum network) is represented by a positive semidefinite process matrix $W$ subject to nested trace conditions that are all linear.

**Definition 29** ($N$-comb [9, 10])**.** An $N$-comb on systems $A_1B_1 \cdots A_NB_N$ is a PSD matrix $W \in \mathrm{PSD}_{d_{A_1}d_{B_1}\cdots d_{A_N}d_{B_N}}$ satisfying the CDP hierarchy of trace conditions. Define the marginals recursively: $W^{(N)} := W$, and for $k = N, N-1, \ldots, 2$:

$$\mathrm{Tr}_{B_k}\big(W^{(k)}\big) = I_{A_k} \otimes W^{(k-1)},$$

with the base condition $\mathrm{Tr}_{B_1}\big(W^{(1)}\big) = I_{A_1}$ (trace-preservation at the first slot). For $N = 1$, this is a single condition $\mathrm{Tr}_{B_1}(W) = I_{A_1}$, recovering a CPTP channel. For $N = 2$, $W$ is a *superchannel*: a map from channels to channels, with two levels of trace conditions. Each CDP condition is an affine equality on $W$.

**Corollary 30** (BP for $N$-combs)**.** *Let $W$ be an extreme $N$-comb. The total number $m$ of real-linear constraints from the CDP hierarchy determines the rank bound via* $\mathrm{rank}(W)^2 \le m$*, i.e.* $\mathrm{rank}(W) \le \lfloor\sqrt{m}\rfloor$.

*Proof.* All CDP trace conditions are affine equalities on $W \in \mathrm{PSD}$. Apply Theorem 7. □

For qubits ($d = 2$ at each slot), the hierarchy of BP bounds is:

| $N$ | Process dim | Constraints $m$ | BP: $r \le \lfloor\sqrt{m}\rfloor$ |
|---|---|---|---|
| 1 | 4 | 4 | 2 |
| 2 | 16 | 52 | 7 |
| 3 | 64 | $\sim 820$ | 28 |
| 4 | 256 | $\sim 13100$ | 114 |

The constraint count grows rapidly: for $N = 2$, the condition $\mathrm{Tr}_{B_2}(W) = I_{A_2} \otimes \sigma$ imposes 48 constraints (an $8 \times 8$ Hermitian matrix constrained to the 16-dimensional subspace $\{I_2 \otimes \sigma\}$), plus 4 from $\mathrm{Tr}_{B_1}(\sigma) = I_{A_1}$, giving $m = 52$.

### 13.4 Application: inclusive scattering

Consider a scattering process with $S$-matrix $S\colon \mathcal{H}_{\mathrm{in}} \to \mathcal{H}_{\mathrm{out}}$ satisfying $S^\dagger S = I$ (isometry on the relevant sector), where $\mathcal{H}_{\mathrm{out}} = \mathcal{H}_{\mathrm{det}} \otimes \mathcal{H}_{\mathrm{unobs}}$ decomposes into detected and unobserved particles. The *inclusive scattering channel* is

$$\Phi_S(\rho) \;=\; \mathrm{Tr}_{\mathrm{unobs}}(S\,\rho\,S^\dagger),$$

which is CPTP (since $S$ is an isometry, $\Phi_S$ is a Stinespring dilation) with Kraus rank bounded by Choi's theorem: $r \le d_{\mathrm{in}}$.

Conservation laws (total angular momentum $J_z$, parity $P$, charge, etc.) impose linear constraints on the Choi matrix $J_{\Phi_S}$. If a conserved charge $Q$ satisfies $[Q_{\mathrm{total}}, S] = 0$, then for the inclusive channel one derives constraints of the form $J_{(i,a),(j,b)} = 0$ whenever the corresponding quantum numbers are incompatible (e.g. $J_z^{(i)} + S_z^{(a)} \ne J_z^{(j)} + S_z^{(b)}$). These are linear constraints on the entries of $J$.

By Theorem 7, the total constraint count $m$ (including both the TP condition and conservation law constraints) determines the BP bound:

$$r \;\le\; \lfloor\sqrt{m}\,\rfloor \qquad \text{where } m = m_{\mathrm{TP}} + m_{\mathrm{cons}}.$$

Here $m_{\mathrm{TP}}$ is the number of independent real constraints from $\mathrm{Tr}_{\mathrm{det}}(J) = I$, and $m_{\mathrm{cons}}$ is the number of independent zero-entry constraints from conservation laws. Physically, $r$ bounds the number of distinct scattering pathways that contribute to the detected particle's state.

### 13.5 Scattering example 1: generic $2 \to 2$ with spin

Consider two incoming particles with total input Hilbert space dimension $d_{\text{in}}$, detecting one outgoing particle in a $d_{\text{det}}$-dimensional space. With $k$ conservation laws:

| $d_{\text{in}}$ | $k=0$ | $k=1$ | $k=2$ | $k=3$ | $k=5$ | $k=10$ |
|---|---|---|---|---|---|---|
| 2 | 2 | 2 | 2 | 2 | 3 | 3 |
| 3 | 3 | 3 | 3 | 3 | 3 | 4 |
| 4 | 4 | 4 | 4 | 4 | 4 | 5 |
| 6 | 6 | 6 | 6 | 6 | 6 | 6 |
| 8 | 8 | 8 | 8 | 8 | 8 | 8 |
| 10 | 10 | 10 | 10 | 10 | 10 | 10 |

Values show $r_{\max} = \lfloor \sqrt{d_{\text{in}}^2 + k} \rfloor$, treating $m_{\text{TP}} = d_{\text{in}}^2$ (complex Choi matrix) and $k$ additional constraints from conservation laws. For moderate $k \ll d_{\text{in}}^2$, the bound remains close to $d_{\text{in}}$. Conservation laws have significant impact only when $k$ is comparable to $d_{\text{in}}^2$.

### 13.6 Scattering example 2: electron–electron (Coulomb)

For $e^- + e^- \to e^- + e^-$ (Coulomb scattering), we use a partial-wave decomposition truncated at orbital angular momentum $L \leq L_{\max}$. The basis states are $|L, M_L, S, M_S\rangle$ with $L + S$ even (antisymmetry of identical fermions), $S \in \{0, 1\}$ (singlet/triplet). The inclusive channel traces out one electron, detecting the other's spin ($d_{\text{det}} = 2$).

Conservation of $J_z = M_L + M_S$ and parity $P = (-1)^L$ impose linear constraints on the Choi matrix. The SDP verification (`numerics/scattering_bp.jl`) confirms the BP bound:

| $L_{\max}$ | $d_{\text{in}}$ | $d_{\text{det}}$ | $m_{\text{TP}}$ | $m_{J_z}$ | $m_P$ | $r_{\text{BP}}$ | $r_{\text{obs}}$ |
|---|---|---|---|---|---|---|---|
| 1 | 10 | 2 | 55 | 154 | 10 | 20 | 9 |
| 2 | 15 | 2 | 120 | 357 | 44 | 31 | 14 |
| 3 | 36 | 2 | 666 | 2216 | 114 | 76 | – |
| 4 | 45 | 2 | 1035 | 3515 | 232 | 97 | – |

The SDP uses a real symmetric Choi matrix (the partial-wave $S$-matrix is real by time-reversal symmetry), so $m_{\text{TP}} = d_{\text{in}}(d_{\text{in}} + 1)/2$ and the BP bound is $r(r + 1)/2 \leq m$ (the real symmetric formula from Theorem 6). For $L_{\max} \leq 2$, the SDP is solved numerically and the observed ranks confirm $r_{\text{obs}} \leq r_{\text{BP}}$. For $L_{\max} \geq 3$, the Choi matrix exceeds $70 \times 70$; dashes indicate analytical bounds only.

### 13.7 Scattering example 3: Compton scattering

For $\gamma + e^- \to \gamma + e^-$ (Compton scattering), the photon has helicities $\lambda_\gamma = \pm 1$ and the electron $\lambda_e = \pm\frac{1}{2}$. In the partial-wave (helicity) basis $|J, \lambda_\gamma, \lambda_e\rangle$ with half-integer $J$ up to $J_{\max}$, the inclusive channel traces out the photon and observes the electron spin ($d_{\text{det}} = 2$).

Conservation of total $J_z = \lambda_\gamma + \lambda_e$ block-diagonalises the Choi matrix. The BP bound on the Kraus rank of the detected-electron channel is verified numerically:

| $J_{\max}$ | $d_{\text{in}}$ | $d_{\text{det}}$ | $m_{\text{TP}}$ | $m_{\text{cons}}$ | $r_{\text{BP}}$ | $r_{\text{obs}}$ |
|---|---|---|---|---|---|---|
| 1 | 6 | 2 | 21 | 54 | 11 | 5 |
| 2 | 10 | 2 | 55 | 153 | 19 | 8 |
| 3 | 14 | 2 | 105 | 302 | 28 | 10 |
| 4 | 18 | 2 | 171 | 501 | 36 | 14 |

All four cases are solved numerically using a real symmetric SDP (with $r(r+1)/2 \leq m$ as the BP formula and $m_{\mathrm{TP}} = d_{\mathrm{in}}(d_{\mathrm{in}}+1)/2$). As $J_{\max}$ increases, $d_{\mathrm{in}}$ grows linearly and the BP bound scales as $r \sim d_{\mathrm{in}}$. The observed ranks are significantly below the BP ceiling: helicity conservation block-diagonalises the Choi matrix, reducing the effective number of free parameters.

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